

MLSys 2026 Agent (Track B) Writeup

Overview

- The agent uses the Gemini API to choose scheduling parameters.
- Parameters selected: scheduler (baseline/fuse), connectivity (strict/loose/auto), retain_budget.
- The deterministic scheduler then generates a valid official schedule JSON.

Prompting

- System prompt enforces JSON-only output with a fixed schema.
- User prompt includes the full problem JSON.
- Temperature is low (0.2) for reproducibility.

Architecture

- 1) Parse input JSON into internal problem representation.
- 2) Call Gemini to select scheduling parameters.
- 3) Run the scheduler with chosen parameters.
- 4) Emit strict official schedule JSON (no extra keys).

Notes

- The agent performs a single Gemini call per benchmark.
- If the Gemini call fails, the agent falls back to safe defaults.